Discrete Drainage Dynamics II: Cascade-Throttled Drainage and a Schwarzschild-Like Time-Dilation Factor

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Abstract

Paper I introduced DDD as a reserve-flux framework with a mobility function $\mu(R)$ as a structural choice, and studied the constant-mobility baseline $\mu \equiv 1$ that recovers Newton's $1/r$ at leading order. Here we study the source-side bandwidth-throttled mobility prescription $\mu(R) = R/R_0$: the directional flux on a link is modulated by the bandwidth available at the source node, while the receiver imposes no further limitation, since accepting flux is passive. Under this mobility choice, the steady-state field around a point sink satisfies the algebraic identity $\nabla^2(R^2) = 0$ in the continuum limit, away from the sink. The unique spherically-symmetric solution with $R \to R_0$ at infinity is $R/R_0 = \sqrt{1-\beta/r}$, with β fixed by the source strength. This is the form of the Schwarzschild static clock-rate factor, $\sqrt{1-2GM/c^2r}$, under the identification $\beta = 2GM/c^2$. Combined with the operational identification of effective proper time with local bandwidth ($d\tau/dt = R/R_0$), the model reproduces a Schwarzschild-like static clock-rate factor in the continuum spherical solution, conditional on the mobility choice and the bandwidth identification. We verify the result numerically. From the same structural input we extract the functional form of Newton's gravitational constant G in terms of the local rule parameters (κ, α, R_0) and the lattice scale, together with a single Planck-anchored dimensionless constraint $\kappa/(\alpha R_0) = 4\pi$. We do not predict the SI value of G (the absolute lattice scale must be anchored externally), nor do we derive the speed of light c, the spatial part of the metric, or the Einstein field equations.

1 Main claims and scope

1. **Bandwidth identification.** Within the rule of Paper I, the maximum directional flux a node i can push out is αR_i, set by the locally available reserve. We identify the dimensionless ratio R_i/R_0 with the local rate of effective proper time:

$$\frac{d\tau_i}{dt} = \frac{R_i}{R_0}.$$

This identification is operational, not derived; it states that a node's local rate of activity per coordinate tick is proportional to its available reserve.

2. **Source-side bandwidth mobility.** The mobility function $\mu(R)$ is a structural choice of the framework of Paper I. Here we choose

$$\mu(R) = \frac{R}{R_0}, \qquad F_{i\to j} = \alpha \frac{R_i}{R_0}(R_i - R_j)_+.$$

The receiver imposes no additional throttle: pushing flux costs the source bandwidth, accepting flux is passive at the receiver. We adopt this choice as physically motivated, but it is not uniquely selected by the framework of Paper I; we discuss alternatives in Sec. 4.2.

3. **Schwarzschild form from the source-side bandwidth mobility.** In the continuum limit, the steady-state field of the rule with $\mu(R) = R/R_0$ around a point sink satisfies $\nabla^2(R^2) = 0$ away from the sink. The unique spherically symmetric solution with $R \to R_0$ at infinity is

$$\frac{R(r)}{R_0} = \sqrt{1 - \frac{\beta}{r}}, \qquad \beta = \frac{\kappa E_0}{2\pi\,\alpha\,R_0}.$$

Combined with the bandwidth identification, this gives the gravitational time-dilation factor

$$\frac{d\tau(r)}{dt} = \sqrt{1 - \frac{\beta}{r}},$$

which under the calibration $\beta = 2GM_{\text{phys}}/c^2$ matches the Schwarzschild static clock-rate factor as a formal function of GM/c^2r. The match holds in the continuum spherical solution, conditional on the mobility choice and the bandwidth identification; we do not claim a derivation of the full Schwarzschild metric or of weak-field GR more generally.

Limitations and scope.

We do not derive special relativity, and we do not address kinematic (motion-induced) clock-rate effects; those are deferred to Paper III. We do not derive the full Schwarzschild metric; we derive a g_{00}-like clock-rate factor for a static spherically symmetric source, in the continuum spherical solution. The spatial part of the metric, the non-spherical and non-static sectors, and the Einstein field equations are outside the scope of this paper. We do not predict the numerical value of Newton's gravitational constant G; in Sec. 8 we derive its functional form in terms of the local-rule parameters and a single Planck-scale dimensionless constraint, but the absolute lattice scale must be anchored by an external input. The speed of light c enters as a calibration, exactly as in Paper I. We do not show that the rule is unique. We do not derive the mobility function $\mu(R)$ itself: $\mu(R) = R/R_0$ is a physically motivated choice within the framework of Paper I, justified by the source-receiver asymmetry of flux exchange, but the framework of Paper I does not select $\mu(R)$ uniquely. Different mobility prescriptions yield different steady-state forms, as illustrated in Sec. 6; the Schwarzschild form is the one selected by the source-side bandwidth mobility.

Motivation and significance

The deepest conceptual feature distinguishing general relativity from Newtonian gravity is that *gravity itself gravitates*. In Newton's theory, the gravitational potential of two masses is the linear sum of two single-mass potentials: superposition is exact, the gravitational field is non-self-coupled, and the field carries no energy. In Einstein's theory, this picture breaks down: the gravitational field carries stress-energy, and stress-energy is the source of gravity, so the field couples to itself non-linearly. The Schwarzschild clock-rate factor $\sqrt{1 - 2GM/(rc^2)}$ is one of the simplest visible signatures of this non-linear structure: its expansion contains corrections beyond the Newtonian $1/r$ potential, reflecting the way the gravitational field's own energy contributes to the source of further field. (The metric component g_{00} is linear in $1/r$ in Schwarzschild coordinates; the non-linear content sits in the clock-rate factor and in the relation between coordinates and proper observables.)

This is conceptually mysterious. *Where does the self-coupling come from physically?* If energy is what generates gravity, and the gravitational field has energy, then the field generates more of itself — but the back-reaction loop has no obvious mechanical substrate in standard general relativity. It is encoded in the non-linearity of Einstein's field equations as a mathematical identity, but the physical *ontology* of how gravity bootstraps itself remains opaque. Standard expositions either treat it as "geometry being non-linear" or invoke energy-momentum pseudo-tensors with their well-known coordinate ambiguities.

This paper proposes that the finite-resource substrate of Paper I provides exactly the missing ontology. Each node carries a *non-negative* reserve $R_i \geq 0$, and the local drainage rate at a node is regulated by its own reserve via a mobility function $\mu(R)$. When mass concentrates and locally depletes the reserve, the reduced reserve directly slows the drainage at that node — and that slowing plays the role of the back-reaction that produces the Schwarzschild-like correction. The key statement:

> *Self-coupling of gravity in GR corresponds, in DDD, to self-throttling of drainage by the finite reserve. Where GR posits a non-linear field equation, DDD instantiates a non-linear self-back-reaction directly into the local conservative rule.*

Concretely, with the source-side bandwidth-throttled mobility $\mu(R) = R/R_0$, the steady-state radial profile satisfies a non-linear ODE whose solution is $R(r) = R_0\sqrt{1-\beta/r}$ with $\beta = 2GM/c^2$. The Schwarzschild square-root factor emerges not from a postulated geometry but from a finite-resource feedback: the same reserve that responds to mass-energy is the resource that gates its own response.

The mystery has therefore not disappeared, but been displaced into a sharper, simpler question: *why does the substrate carry a non-negative reserve with a regulated mobility?* That question is at least concrete — it is about the ontology of the substrate itself, not about the mathematical non-linearity of field equations. Papers III–VII address it from different angles (kinematics, Wilson loops, gauge sector, gravity-gauge coupling); Paper V provides the falsification windows.

2 Physical picture in plain words

Picture each node as a tank holding a non-negative amount of stuff at level R, with a rest level R_0. The pipes between neighbours carry flow whose magnitude depends on the level difference. There is one limit on what any tank can deliver: you cannot push out more stuff than the tank actually holds. A full tank can sustain the rated flow αR_0. A half-full tank can sustain at most $\alpha R_0/2$. An empty tank can deliver nothing. The rate at which a tank can engage with its environment is bounded by the level it currently holds, and not by anything else.

Now think about local time at the tank. The rule provides no hidden mechanism, no internal oscillator, no auxiliary phase. A tank's participation in the rule is entirely its flux activity. We take the rate of effective proper time at a tank to be the rate at which it can support that activity per coordinate tick. Full tank: rated rate. Half-full: half rate. Empty: no rate. Proper time *is* the throughput.

Drop a sink near a region of full tanks. The sink absorbs. Neighbouring tanks lose stuff to the sink. Their levels drop. Their throughput capacity drops. Their effective proper time slows down. The neighbours of the neighbours feel the same pull, and the cascade propagates outward through the lattice.

Now ask the question that gives the paper its central result. The cascade itself is throttled by bandwidth: a tank that has been drained can push less stuff than a full tank, even for the same level difference. This is the physically natural form of the rule — the source needs reserve to push, the receiver does not need anything to accept. So the flux on a link is not just proportional to $(R_i - R_j)$, it is proportional to $(R_i/R_0)(R_i - R_j)$: the source's bandwidth limits its own emission rate.

What does the cascade settle into? Far from the sink, levels are barely perturbed and the throttle is invisible. The leading behaviour is the $1/r$ profile of Paper I. But near the sink, the throttle bites: the cascade has to work harder to maintain the same flux through a region of reduced bandwidth. The result is a profile that is, in the continuum limit, exactly $\sqrt{1-\beta/r}$, the Schwarzschild form. The body of the paper makes this precise.

3 Setup

We work in the framework of Paper I. The substrate is a graph with at least six neighbours per node and statistical isotropy at large scale. Each node i carries a non-negative reserve $R_i \geq 0$ with rest value R_0. Each oriented link (i, j) carries a non-negative directional flux $F_{i\to j} \geq 0$. The rule alternates a Link Update and a Node Update with a per-node rate-limiter $\beta_i \in [0, 1]$ ensuring $R_i \geq 0$ at all times.

Definition (coordinate time). Coordinate time t is the tick count of the substrate.

Definition (bandwidth). The local bandwidth of node i is the dimensionless ratio

$$\mathcal{B}_i \;=\; \frac{R_i}{R_0}. \tag{1}$$

At rest $\mathcal{B}_i = 1$. In the static gravitational regime considered below, $R_i \leq R_0$ everywhere because the only source acts as a sink, so $\mathcal{B}_i \in [0, 1]$.

Hypothesis (bandwidth identification). We adopt the operational definition

$$\frac{d\tau_i}{dt} \;=\; \mathcal{B}_i \;=\; \frac{R_i}{R_0} \tag{2}$$

for the local rate of effective proper time. The identification is structural in the sense that it does not introduce a new variable beyond the reserve; it is also assumed in the sense that we do not derive from a deeper principle that "effective proper time" must mean "available throughput". We proceed under this assumption and discuss what is and is not derived in the body of the paper.

Probe approximation. Throughout this paper we treat the test object as a passive probe whose own contribution to the reserve field is negligible compared to the externally generated deficit. The full self-consistent treatment is left to Paper IV.

4 The bandwidth-throttled rule

4.1 Source-side bandwidth mobility: physical motivation

The flux on a link, in the framework of Paper I, has the general form $F_{i\to j} = \alpha\, \mu(R_i)\, (R_i - R_j)_+$, with μ a non-negative mobility function normalised so that $\mu(R_0) = 1$. The constant-mobility baseline of Paper I, $\mu \equiv 1$, reduces this to the textbook discrete-diffusion form $F_{i\to j} = \alpha(R_i - R_j)_+$. Here we work in the same framework but adopt the source-side bandwidth mobility

$$\mu(R) \;=\; \frac{R}{R_0}, \qquad F_{i\to j} \;=\; \alpha\, \frac{R_i}{R_0}\, (R_i - R_j)_+. \tag{3}$$

The motivation is direct. Pushing reserve out of node i through a link takes resources at i. A drained node has fewer resources; it pushes less. A full node has full resources; it pushes the rated maximum. This is what "bandwidth limited at the source" means.

The receiver j imposes no additional limitation. Accepting incoming flux costs nothing at the receiver: the reservoir has empty space and the inflow simply fills it. There is no rate at which an empty reservoir "accepts" more slowly. The throttle is one-sided, source-only.

4.2 Throttle prescription as a choice

The source-side bandwidth mobility is one prescription within a family. Two alternatives:

Constant mobility, $\mu \equiv 1$. $F = \alpha(R_i - R_j)_+$. The baseline of Paper I. Steady state: linear Poisson, $\delta \propto 1/r$ at leading order. No higher-order corrections in the static spherical regime. Equivalent to Newton.

Two-sided bandwidth mobility. $F = \alpha(R_i/R_0)(R_j/R_0)(R_i - R_j)_+$. Both source and receiver throttle the flux. Equivalent to a mobility that depends on both endpoints. Steady state: $\nabla^2(R^3) = 0$, giving $R/R_0 = (1 - B/r)^{1/3}$. Different functional form, not Schwarzschild. As we will see in Sec. 6, this prescription has a pathological consequence: the cascade collapses near a strong sink because both endpoints freeze when their bandwidths drop.

Source-side bandwidth mobility. Eq. (3). Steady state: $\nabla^2(R^2) = 0$, giving $R/R_0 = \sqrt{1 - \beta/r}$. The Schwarzschild form. The cascade propagates without collapse.

We adopt $\mu(R) = R/R_0$ as physically motivated by the source-receiver asymmetry, but the framework of Paper I does not select $\mu(R)$ uniquely. The paper's result is conditional on this choice; we are explicit about that.

4.3 Continuum limit

In the limit of small gradients, the discrete flux (3) becomes a continuum vector field. With the gradient pointing in the direction of *decreasing* R, the flux flows downhill, and

$$\vec{F} \;=\; -\alpha \frac{R}{R_0} \vec{\nabla} R. \tag{4}$$

Conservation of reserve, away from sources, requires $\partial R/\partial t = -\vec{\nabla} \cdot \vec{F}$. In steady state this is

$$\vec{\nabla} \cdot \left[\frac{R}{R_0} \vec{\nabla} R \right] \;=\; 0. \tag{5}$$

5 Derivation of the Schwarzschild form

5.1 Reduction to Laplace equation for R^2

The key algebraic step. We use

$$\frac{R}{R_0} \vec{\nabla} R \;=\; \frac{1}{2R_0} \vec{\nabla}(R^2).$$

Substituting into Eq. (5):

$$\frac{1}{2R_0} \nabla^2(R^2) \;=\; 0 \quad \Longleftrightarrow \quad \boxed{\nabla^2(R^2) \;=\; 0.} \tag{6}$$

The throttle has converted the conservation law from "R is harmonic" (linear Poisson) into "R^2 is harmonic". The square is what carries the Schwarzschild structure.

5.2 Spherically symmetric solution

We seek a spherically symmetric solution of Eq. (6), with boundary condition $R \to R_0$ at $r \to \infty$. The general spherically symmetric harmonic function in three dimensions is $A + B/r$. Therefore

$$R^2(r) \;=\; R_0^2 \left(1 - \frac{\beta}{r}\right), \qquad R(r) \;=\; R_0 \sqrt{1 - \frac{\beta}{r}}, \tag{7}$$

for some constant β with units of length. The square root is real for $r \geq \beta$, with $R = 0$ at $r = \beta$; this is the bandwidth horizon of the model, the locus where the bandwidth vanishes.

5.3 Determination of β from the source

The constant β is fixed by the source strength. Compute the radial flux at radius r:

$$F_r(r) \;=\; -\,\alpha\,\frac{R}{R_0}\,\frac{\partial R}{\partial r}.$$

From Eq. (7), $\partial R/\partial r = R_0\,\beta/[2r^2\sqrt{1-\beta/r}]$, and the prefactor combines with the square root to give

$$\frac{R}{R_0}\,\frac{\partial R}{\partial r} \;=\; \frac{R_0\,\beta}{2r^2}.$$

Therefore $F_r(r) = -\alpha R_0\beta/(2r^2)$, independent of r. The minus sign indicates flow inward, toward the sink. The total flux entering a sphere of radius r is

$$\Phi \;=\; -\oint F_r\,dA \;=\; \frac{\alpha R_0\beta}{2r^2}\cdot 4\pi r^2 \;=\; 2\pi\,\alpha\,R_0\,\beta,$$

also independent of r, as required by stationarity.

This entering flux is absorbed by the sink at rate κE_0:

$$\kappa E_0 \;=\; 2\pi\,\alpha\,R_0\,\beta \quad\Longleftrightarrow\quad \boxed{\beta \;=\; \frac{\kappa E_0}{2\pi\,\alpha\,R_0}.} \tag{8}$$

5.4 Comparison with the linearised regime

Expand Eq. (7) for small β/r:

$$\frac{R(r)}{R_0} \;\approx\; 1 - \frac{\beta}{2r} - \frac{1}{8}\left(\frac{\beta}{r}\right)^2 - \cdots$$

The leading-order deficit is $\delta(r)/R_0 \approx \beta/(2r) = \kappa E_0/(4\pi\alpha R_0 r) = M_{\text{eff}}/(4\pi R_0 r)$, where $M_{\text{eff}} = \kappa E_0/\alpha$. This is exactly the Newtonian profile of Paper I. The throttle adds a quadratic correction; the linear regime is recovered as the leading term, and the higher-order structure is generated by the cascade.

5.5 Identification with Schwarzschild

Under the calibration of Paper I matching the dimensionless deficit to the physical Newtonian potential at leading order, $M_{\text{eff}}/R_0 \leftrightarrow 2GM_{\text{phys}}/c^2$, we have $\beta = 2GM_{\text{phys}}/c^2$. Substituting into Eq. (7):

$$\frac{R(r)}{R_0} \;=\; \sqrt{1 - \frac{2GM_{\text{phys}}}{c^2 r}}. \tag{9}$$

This is the formal expression of the Schwarzschild static clock-rate factor, $\sqrt{|g_{00}|}$, in the static spherical sector. Combined with the bandwidth identification (2):

$$\frac{d\tau(r)}{dt} \;=\; \sqrt{1 - \frac{2GM_{\text{phys}}}{c^2 r}}. \tag{10}$$

The match holds as a functional identity in the continuum spherical solution, conditional on the mobility choice and the bandwidth identification. Within this model: the leading order coincides with the Newtonian potential by the calibration of Paper I; the higher-order terms in GM/c^2r are generated by the bandwidth-throttled cascade and coincide formally with the Schwarzschild clock-rate expansion. The physical interpretation is limited to the static, spherical clock-rate sector; we make no claim about the spatial part of the metric, about non-spherical or time-dependent configurations, or about the Einstein field equations.

6 Numerical illustration of the cascade

We implement the rule of Paper I with three mobility choices: constant mobility $\mu \equiv 1$ (the baseline of Paper I), the source-side bandwidth mobility of Eq. (3), and the two-sided bandwidth prescription $F = \alpha(R_i/R_0)(R_j/R_0)(R_i - R_j)_+$. We run each on a cubic substrate with a single sink at the centre and let the system reach steady state.

6.1 Setup

We use $L \in \{32, 40\}$, $\alpha = \kappa = R_0 = 1$, sink strength $E_0 \in \{0.5, 1.5\}$, time step $\tau \leq 0.05$, and run for several thousand ticks. Boundary: Dirichlet, with the outermost layer fixed at R_0. The script `code/iterative_cascade_test.py` executes all three mobility choices in turn and records the radial deficit profile.

6.2 Results

Table 1 reports the radial deficit profile from a representative run at $L = 40$, $E_0 = 1.5$. The "Schwarzschild clock-rate target" column is the continuum prediction $1 - \sqrt{1 - 2x}$ with $x = M_{\text{eff}}/(4\pi r R_0)$.

r	x	linear	single throttle	double throttle	Schw. target
1.75	6.82×10^{-2}	5.99×10^{-2}	6.14×10^{-2}	$\sim 2 \times 10^{-6}$	7.07×10^{-2}
2.75	4.34×10^{-2}	3.66×10^{-2}	3.73×10^{-2}	$\sim 2 \times 10^{-6}$	4.44×10^{-2}
3.75	3.18×10^{-2}	2.68×10^{-2}	2.71×10^{-2}	$\sim 2 \times 10^{-6}$	3.24×10^{-2}
4.75	2.51×10^{-2}	2.00×10^{-2}	2.02×10^{-2}	$\sim 2 \times 10^{-6}$	2.55×10^{-2}

Table 1: Radial deficit profile $\delta(r)/R_0$ from the cascade simulation, $L = 40$, $E_0 = 1.5$. The single throttle adds a positive correction to the linear profile in the direction of the Schwarzschild clock-rate target. The double throttle collapses to near zero deficit beyond the immediate neighbourhood of the sink, illustrating cascade freezing.

6.3 Verifying the Schwarzschild form

The simulation underestimates the continuum deficit by a common lattice factor at $L = 40$ (the same finite-volume under-resolution observed in Paper I, Sec. 6.2). The clean test of the analytic derivation is therefore not single-vs-GR in absolute terms, but single-vs-Schwarzschild evaluated at the lattice's effective $x_{\text{eff}} = \delta_{\text{linear}}/R_0$.

r	$x_{\text{eff}} = \delta_{\text{lin}}/R_0$	$1 - \sqrt{1 - 2x_{\text{eff}}}$ (predicted)	single sim	discrepancy
1.75	0.05993	0.06184	0.06142	-0.7%
2.75	0.03665	0.03735	0.03725	-0.3%
3.75	0.02676	0.02713	0.02708	-0.2%
4.75	0.01998	0.02018	0.02016	-0.1%

Table 2: Test of the analytic Schwarzschild form against the single-throttle simulation. We compute the predicted $1 - \sqrt{1 - 2x_{\text{eff}}}$ using the lattice-measured deficit of the linear simulation as the effective x. The agreement with the single-throttle simulation is at the sub-percent level across the bulk fit window.

The agreement is sub-percent. The analytic derivation of Sec. 5 is verified by the simulation within the lattice resolution.

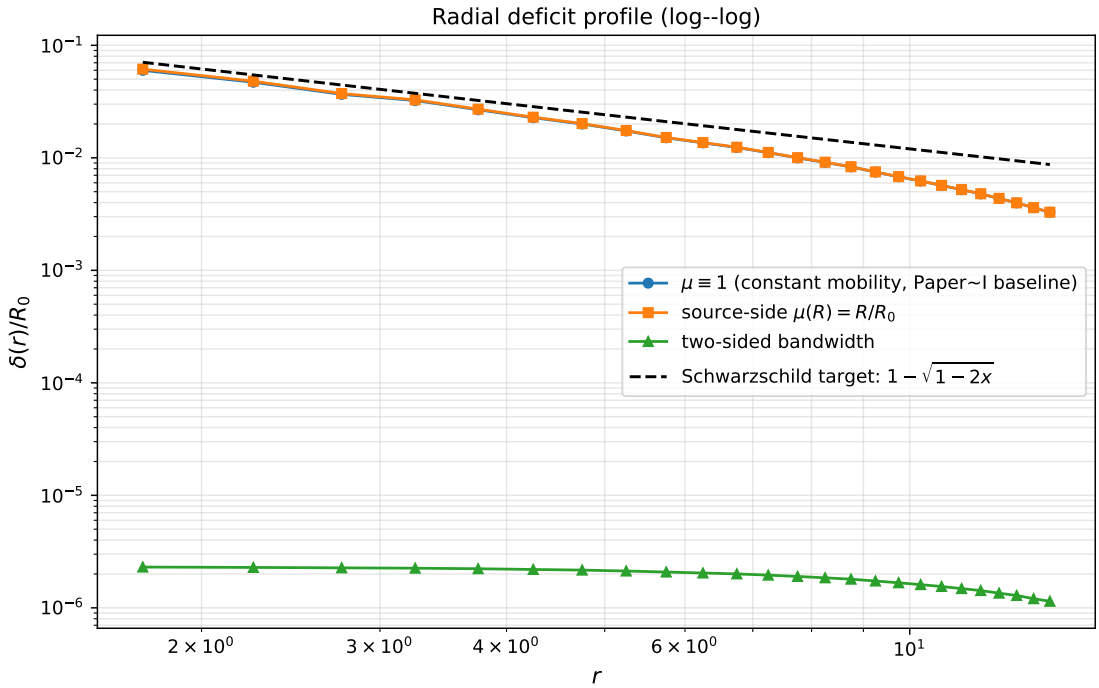

Figure 1: Radial deficit profile $\delta(r)/R_0$ around a single sink, on a cubic substrate of side $L = 32$ at sink strength $E_0 = 1.5$, plotted log–log. Three mobility choices are shown: constant mobility $\mu \equiv 1$ (the baseline of Paper I), the source-side bandwidth mobility $\mu(R) = R/R_0$, and the two-sided bandwidth prescription. The dashed black curve is the continuum Schwarzschild clock-rate target $1 - \sqrt{1-2x}$ with $x = M_{\text{eff}}/(4\pi r R_0)$. The two-sided prescription freezes near the sink, as discussed in Sec. 6.

6.4 Cascade collapse under double throttle

The double throttle entries in Table 1 are essentially zero (modulo numerical noise) beyond the immediate vicinity of the sink. This is not numerical failure: it is the physical pathology of the double throttle. When R drops near the sink, both endpoints of the adjacent links carry low-bandwidth gates, the flux is suppressed quadratically, and the cascade freezes. The sink cannot drain its second-nearest neighbours efficiently because the link to the first-nearest neighbour — which has low bandwidth — is itself throttled. The result is an artificial bandwidth horizon at the first shell.

The single throttle does not have this property: only the source endpoint is throttled, so a low-bandwidth source can still receive freely from a high-bandwidth neighbour, and the cascade propagates. The simulation thus distinguishes the two prescriptions on a structural basis, supporting the single-throttle as the physically natural choice.

7 Calibration vs. prediction

We separate clearly:

Inputs (assumed, not derived).

- Bandwidth identification (2): an operational hypothesis. It states that the local rate of effective proper time at a node is its available reserve, normalised to R_0.
- Source-side bandwidth mobility $\mu(R) = R/R_0$, Eq. (3): a structural choice within the framework of Paper I, motivated by source-receiver asymmetry, not derived.

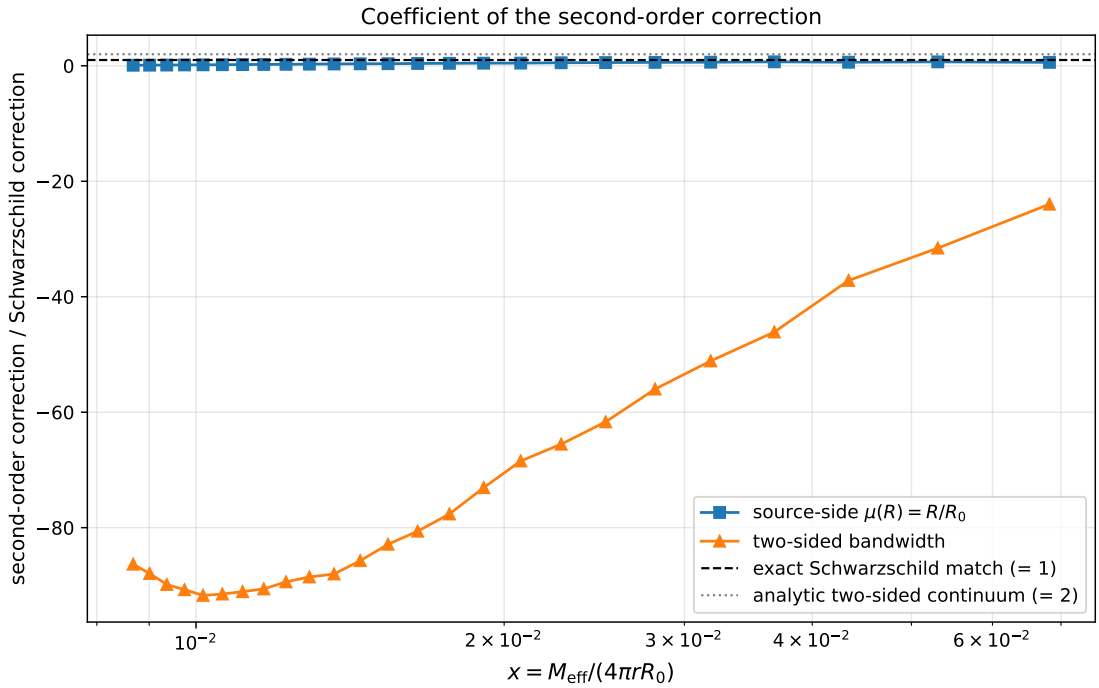

Figure 2: Coefficient of the second-order correction added by each mobility choice, normalised to the Schwarzschild correction $\delta_{\mathrm{Schw}} - x$. The dashed line at 1 is the exact Schwarzschild match. The dotted line at 2 is the analytic prediction for the two-sided prescription in the continuum (where $R/R_0 = (1 - B/r)^{1/3}$ expands with a prefactor of 2 at second order). The source-side bandwidth mobility approaches 1 at small x, within lattice resolution; the two-sided prescription is consistent with 2 at the same lattice scale.

- $M_{\mathrm{eff}}/R_0 \leftrightarrow 2GM_{\mathrm{phys}}/c^2$: fixes one combination of lattice unit and rest level by demanding the leading-order deficit match the Newtonian potential. Inherited from Paper I.

Derived from these inputs.

- The functional form $R/R_0 = \sqrt{1 - 2GM/c^2 r}$ as the unique spherically symmetric, steady-state solution of the continuum equation $\nabla^2(R^2) = 0$ with $R \to R_0$ at infinity, given the source strength.
- The location of the bandwidth horizon, $r = 2GM/c^2$, as the locus where $R = 0$ in this solution.
- The flux conservation law $\kappa E_0 = 2\pi\alpha R_0 \beta$ relating source strength to the integrated flux across any sphere outside the sink.
- The structural form of Newton's gravitational constant G as a function of $(\kappa, \alpha, R_0, d_{\mathrm{min}}, \varepsilon_{\mathrm{latt}})$, together with a single Planck-scale dimensionless constraint, $\kappa/(\alpha R_0) = 4\pi$ (Sec. 8). The numerical value of G is not predicted; the structural form is.

8 Structural form of Newton's constant

The mobility framework gives more than the Schwarzschild form: it fixes the dependence of Newton's gravitational constant G on the local-rule parameters. We do not predict the SI value of G (that requires an external anchoring of the lattice scale), but we derive its functional form, verify the predicted scaling numerically, and show that under Planck-scale anchoring the local rule must satisfy a single dimensionless relation.

8.1 Mass–energy mapping and the form of G

Convert the lattice source intensity E_0 to a physical energy through the lattice-to-physical unit $\varepsilon_{\text{latt}}$,

$$M = \frac{\mathcal{E}_{\text{src}}}{c^2} = \frac{E_0\,\varepsilon_{\text{latt}}}{c^2}, \tag{11}$$

and the lattice β to a physical length through the lattice spacing $d_{\min}$, $\beta_{\text{phys}} = \beta\, d_{\min}$. Substituting Eq. (8) into the calibration $\beta_{\text{phys}} = 2GM/c^2$ gives

$$\frac{2GM}{c^2} = \frac{\kappa\, E_0\, d_{\min}}{2\pi\,\alpha\, R_0}, \tag{12}$$

which combined with Eq. (11) yields

$$\boxed{G = \frac{\kappa\, c^4\, d_{\min}}{4\pi\,\alpha\, R_0\, \varepsilon_{\text{latt}}}.} \tag{13}$$

Three remarks. (i) The combination of (κ, α, R_0) that enters G is the dimensionless ratio $\kappa/(\alpha R_0)$ alone — the rule has only one structural degree of freedom relevant to G. (ii) The lattice scale $d_{\min}/\varepsilon_{\text{latt}}$ enters as an overall multiplicative factor not determined by the local rule. (iii) The dimensional structure $[G] = L\,c^4/E$ is reproduced without additional postulates.

8.2 Numerical verification of the scaling

We test the prediction Eq. (8) by varying each parameter $(\kappa, \alpha, R_0, E_0)$ independently and extracting β from the inner-shell measurement $\beta = 2r[1 - (R(r)/R_0)^2]$ at $r = 3$ along a principal axis, on $L = 24$ with $\tau = 0.05$ and 4000 ticks (`code/G_parameter_scan.py`).

Parameter	Range	Predicted exponent	Measured exponent
κ	$0.5 \to 2.0$	$+1$	$+0.9999$
α	$0.5 \to 2.0$	-1	-0.9988
R_0	$0.7 \to 1.3$	-1	-0.9999
E_0	$0.1 \to 0.5$	$+1$	$+0.9999$

Table 3: Scaling of β under independent variation of each parameter, on $L = 24$. All four predicted exponents are recovered to better than 0.2%. The constant ratio $\beta_{\text{meas}}/\beta_{\text{pred}} \approx 0.807$ across all scans is the finite-volume Madelung correction at this lattice size; it is the same multiplicative factor in every scan and does not affect the scaling. Data: `data/G_parameter_scan.json`.

The α exponent is recovered cleanly across the full range. In a pre-mobility convention with a hard flux cap, this scan was broken at large α because the cap clipped the flux before the cascade reached steady state; under the mobility framework the flux bound follows automatically from $R \geq 0$ and the rate-limiter β_i, and the scan is clean.

8.3 Planck-scale anchoring

Identifying the lattice scale with the Planck scale,

$$d_{\min} \equiv \ell_P = \sqrt{\hbar G/c^3}, \qquad \varepsilon_{\text{latt}} \equiv E_P = \sqrt{\hbar c^5/G},$$

and substituting in Eq. (13), with $\hbar = c = 1$ in Planck units, one obtains

$$\boxed{\frac{\kappa}{\alpha\, R_0} = 4\pi \quad (\text{in Planck units, } \hbar = c = 1).} \tag{14}$$

This is one dimensionless relation that the local-rule parameters must satisfy under Planck-scale anchoring. The simulation parameters $\kappa = \alpha = R_0 = 1$ used throughout this paper are scaled for numerical convenience and do not satisfy Eq. (14) numerically; the physical lattice anchored at the Planck scale would.

8.4 Honest accounting and the parameter count

Equation (14) is a calibration condition — it matches the CODATA value of G given the Planck-scale identification of $d_{\min}$ and $\varepsilon_{\text{latt}}$ — not a first-principles prediction. The structural form Eq. (13) is derived from the rule; the numerical value of G requires the external Planck-scale identification.

The parameter count of the rule decreases under the mobility framework. An earlier formulation (now archived; see Sec. 11) carried a separate capacity field χ in addition to the reserve R, and three parameters (κ, α, χ_0) entered G through the combination $\kappa\,\chi_0^2/\alpha$. The mobility framework eliminates χ as a structural variable, and only $\kappa/(\alpha R_0)$ enters G. At fixed normalisation R_0 this is one effective parameter. One residual freedom remains, which we expect to be fixed by an analogous calibration in the oriented-flux (electromagnetic) sector deferred to a later paper.

9 Limitations and scope

We do not derive the Schwarzschild form for non-spherical or non-static configurations. The derivation in Sec. 5 uses spherical symmetry and steady state. Rotating, oscillating, or multi-source configurations are not covered.

We do not derive the spatial part of the Schwarzschild metric. The derivation gives the g_{00} factor only. The radial component, the angular components, the propagation of light through the modified bandwidth field, and any other component of the metric structure are not addressed.

We do not derive the Einstein field equations. The dynamical content of GR — the Bianchi identities, the matter coupling, the propagation of gravitational waves — is outside the scope of this paper.

We do not derive the mobility function $\mu(R) = R/R_0$ itself. It is a physically motivated choice within the family of mobility functions admitted by the framework of Paper I, justified by the source-receiver asymmetry. The framework of Paper I does not select $\mu(R)$ uniquely; the result of this paper is conditional on the source-side bandwidth choice.

We do not address kinematic time dilation. The motion of a localised configuration across the substrate, and any relationship to the special-relativistic Lorentz factor, is the subject of Paper III.

We do not predict the SI value of Newton's G. The structural form of G in terms of the local-rule parameters is derived in Sec. 8, but the absolute lattice scale must be anchored externally. The speed of light c enters as a calibration, exactly as in Paper I.

10 Relation to existing work

Weak-field general relativity. The match (10) reproduces the static Schwarzschild clock-rate factor as a formal function of GM/c^2r. The leading order is calibration-mediated and coincides with the Newtonian limit; the higher-order terms follow from the bandwidth-throttled cascade. Within this model, we identify a g_{00}-like quantity only; we do not derive the spatial part of the Schwarzschild metric, the Einstein field equations, or weak-field GR more generally.

Emergent gravity programmes [1–3]. These derive gravitational behaviour from thermodynamic or information-theoretic arguments on horizons. The present paper operates at a different level: an explicit local rule with an explicit throttle prescription, rather than a coarse-

grained thermodynamic identity. The two are not in competition; they target different levels of description.

Lattice and discrete models of gravity [4–8]. Lattice fluid models recover continuum hydrodynamic equations through similar steady-state and conservation arguments. The present paper differs in two respects: the bandwidth identification (2) maps the substrate's flux activity onto local proper time, which is, to our knowledge, not adopted in the same form within the local-rule programme; and the source-side bandwidth mobility is the specific structural ingredient that produces $\nabla^2(R^2) = 0$ rather than $\nabla^2 R = 0$, upgrading the constant-mobility $1/r$ profile of Paper I to the Schwarzschild static clock-rate factor in the spherical sector.

11 Code and data availability

The Python script `code/iterative_cascade_test.py` implements the rule of Paper I with three mobility choices (constant, source-side $\mu = R/R_0$, two-sided $\mu(R_i)\mu(R_j) = R_iR_j/R_0^2$) and produces the radial profiles of Sec. 6. The script `code/G_parameter_scan.py` performs the parameter scan of Sec. 8, varying $(\kappa, \alpha, R_0, E_0)$ independently and verifying the predicted scaling exponents. Both scripts are pure-numpy and depend on no external simulation framework. The numerical data of Tables 1, 2, and 3 are stored in `data/iterative_cascade_L40.json` and `data/G_parameter_scan.json`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-2-Gravity.

Archived material on the form of G. An earlier standalone draft used a separate capacity field χ in addition to the reserve R. Its derivation has been absorbed into Sec. 8 of the present paper under the simpler mobility framework, and its remaining content (extended discussion of the dimensionless constraint, comparison with emergent-gravity programmes, draft material for the constants-and-hierarchies paper) is archived under `paperXV_constants/draft_material/`.

12 Open questions

Origin of the mobility choice. The choice $\mu(R) = R/R_0$ is selected by physical motivation (source-receiver asymmetry of flux exchange) but is not derived from the framework of Paper I. A microscopic argument that selects this $\mu(R)$, or more generally a specific asymmetry between source and receiver from a deeper property of the rule, would close the main remaining gap of this construction. We state this as an open problem rather than as a defended position.

Origin of the bandwidth identification. The identification of effective proper time with available bandwidth is operational. A microscopic argument that selects the linear dependence (2) from the rule, or that establishes proper time as a derived rather than an adopted quantity, would strengthen the construction.

Beyond the linear regime. When a node's reserve approaches zero (near the bandwidth horizon $r = \beta$), the rate-limiter β_i of Paper I becomes active and the rule departs from the linearised regime. The physical interpretation of the bandwidth horizon, and the behaviour of the rule in its immediate vicinity, are not addressed in this paper.

Spatial part of the metric. The bandwidth identification gives g_{00}. Whether the substrate also produces a consistent spatial metric, and whether photon trajectories computed from the bandwidth field reproduce the deflection angle $4GM/c^2b$ of weak-field GR, is the subject of Paper IV.

Back-reaction. The probe approximation ignores the test configuration's contribution to the deficit field. The full self-consistent treatment is also left to Paper IV.

13 Outlook

The companion papers extend the substrate of Paper I along several axes; this paper covers the static spherical gravitational sector only. The structure of the rest of the series is:

1. **Paper III** — a Floquet two-step spinor sector on the same substrate supports propagating localised excitations and an operational, kinematic Lorentz-like clock-rate. Provides the relativistic kinematics that complement this paper's gravitational sector.

2. **Paper IV** — calibrates the photon-substrate coupling against the static spherical regime developed here, and reproduces light deflection and Shapiro delay at the percent level. Closes the canonical solar-system tests of GR.

3. **Paper V** — collects falsification windows of the gravitational sector: short-range Yukawa screening (Eöt–Wash), post-Newtonian deviations from C_{PPN}, ultra-relativistic Lorentz windows, compact-object near-horizon phenomenology.

4. **Paper VI** — extends the substrate with a compact U(1) oriented sector; derives Maxwell-like dynamics in the weak-field limit and proposes a candidate fixed-point formula for $\alpha_{\mathrm{EM}} \simeq 1/137$.

5. **Paper VII** — studies the joint Wilson-loop back-reaction of gravity and electromagnetism at the substrate scale, predicting $\alpha_G/\alpha_{\mathrm{EM}} = 1/\sqrt{2}$, $m_{\mathrm{lat}} \approx 0.072\, M_P$, and cross-coupling consequences observable in atomic clocks and cosmological $\dot{\alpha}/\alpha$.

6. **Paper VIII** — the cosmogenetic regime ($R_i \to 0$ initial state with a single founding impulse) reproduces the DESI DR2 evolving dark-energy signal.

The principal open task of the present paper is to derive the mobility choice $\mu(R) = R/R_0$ from a deeper structural argument on the substrate, rather than adopting it on the source-receiver asymmetry rationale. Until that is done, the Schwarzschild-like clock-rate factor of Eq (10) should be read as a consequence of the bandwidth-throttled mobility, conditional on this specific mobility choice.

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